

Ubiquitous Sensing, Computing & Communication Laboratory

Quality Laboratory Manual

Experiment No. 06

Design and implement a program that detect LPG gas using Gas sensor and Microcontroller.

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Title of Experiment: Design and implement a program that detect LPG gas using Gas sensor and Microcontroller.

Aim of Experiment: To design and implement an LPG gas leakage detection system using an MQ-6 gas sensor and Raspberry Pi, which alerts the user upon detection of hazardous gas levels.

System Requirements – Raspberry pi 4 Board Win 10 and above OS, 4GB RAM, 2.33 GHz Processor

Software/s Needed for Experiment –

Experiment Objectives:

1. To interface the MQ-6 gas sensor with Raspberry Pi using GPIO.
2. To continuously monitor the presence of LPG gas in the environment.
3. To trigger an alert (buzzer or LED) when LPG concentration exceeds a safe threshold.
4. To understand the basic operation of gas sensors and their application in safety systems.

Experiment Outcomes:

1. **Understand the working principle of an MQ-6 gas sensor** and how it detects LPG and other combustible gases.

Interface a gas sensor with Raspberry Pi using GPIO pins for real-time monitoring.

Write Python code to read digital signals from the sensor and trigger alerts when gas leakage is detected.

2. **Implement a basic gas leakage alert system**, which can be extended for real-world safety applications.
3. **Develop awareness about safety measures** in embedded systems and IoT applications related to gas leakage detection.

1.Theory:

LPG (Liquefied Petroleum Gas) is a commonly used fuel for domestic and industrial applications. However, leakage of LPG gas can be hazardous and may lead to fire or explosion. Therefore, detecting LPG leaks at an early stage is crucial to prevent accidents and ensure safety. This experiment uses an MQ-6 gas sensor interfaced with a Raspberry Pi to detect LPG gas and alert the user when the concentration exceeds a predefined threshold.

Working Principle of MQ-6 Sensor

The MQ-6 gas sensor is a semiconductor-type sensor that detects the presence of LPG, butane, propane, and other combustible gases. It contains a SnO_2 (tin dioxide) sensitive layer, which changes its resistance when exposed to gas.

- In clean air, the sensor has high resistance.
- When LPG is present, resistance decreases, allowing more current to flow, which the sensor interprets as increased gas concentration.

The sensor provides:

- Analog output (AO): Proportional to gas concentration.
- Digital output (DO): Goes HIGH or LOW based on an internal comparator and threshold level.

Role of Raspberry Pi

Raspberry Pi is used to:

- Read the digital signal from the MQ-6 sensor via GPIO.
- Monitor for gas presence continuously.
- Trigger a buzzer or LED if leakage is detected.
- Run a Python program that processes the sensor input and takes appropriate action.

2. Wiring Connections:

Interfacing a gas sensor like the MQ-2 with a Raspberry Pi Pico involves connecting the sensor's analog output to a GPIO pin on the Pico and then reading the signal using the Pico's analog-to-digital converter (ADC). A program, typically in MicroPython, is then written to read the ADC value and interpret it to determine gas presence or concentration.

1. Connecting the Gas Sensor:

- **Power and Ground:**

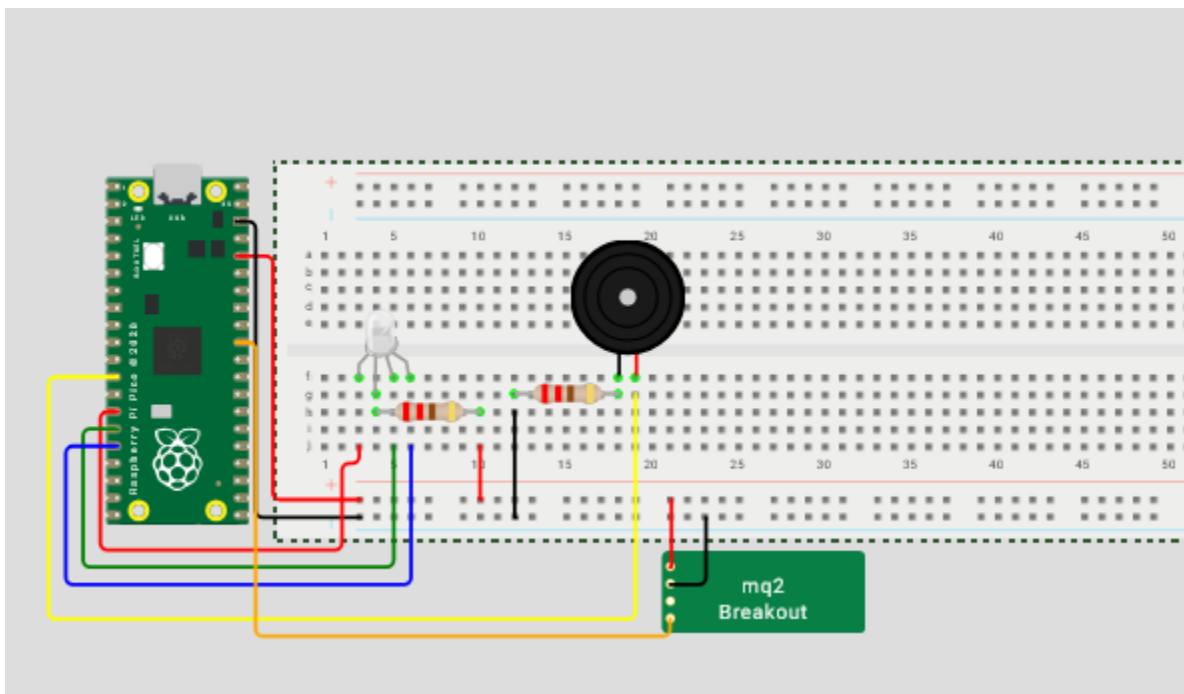
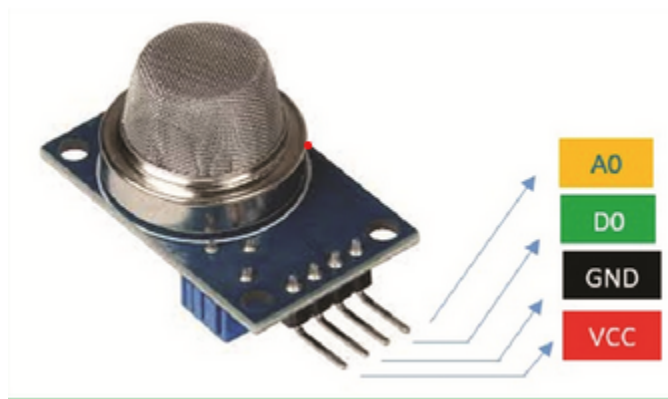
Connect the gas sensor's VCC to the Pico's 3.3V pin and the ground pin to the Pico's ground.

- **Analog Output:**

Connect the gas sensor's analog output to one of the Pico's GPIO pins that can be used for ADC readings (e.g., GPIO 26).

- **Optional:**

You may need a voltage divider circuit to scale the sensor's output voltage to the Pico's ADC input range if necessary.

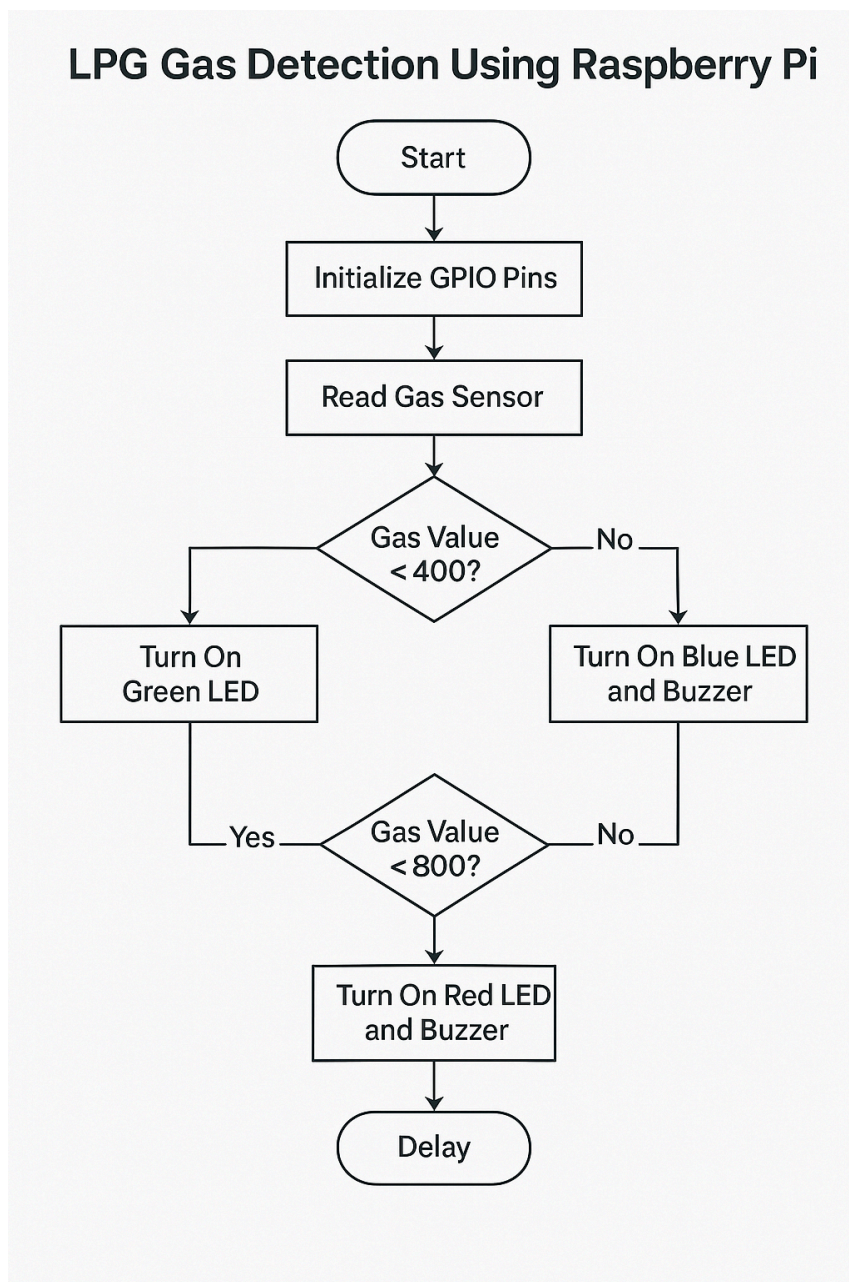


3. Algorithm

Algorithm (Step-by-Step)

1. **Start**
2. **Initialize** the gas sensor, LEDs, and buzzer.
3. **Read** the gas level (analog value) from the gas sensor.
4. **Display** the gas value on the serial monitor.
5. **Check gas level:**
 - If the gas value is **less than 400** → Safe
 - Turn ON Green LED
 - Turn OFF Buzzer
 - If the gas value is **between 400 and 800** → Caution
 - Turn ON Blue LED
 - Turn ON Buzzer with 1kHz beep
 - If the gas value is **more than 800** → Danger
 - Turn ON Red LED
 - Turn ON Buzzer with 2kHz beep
6. Wait for 1 second
7. **Repeat from step 3**

4. Flowchart



5. Program

```
// Pin Definitions
#define GAS_SENSOR_PIN 26
#define RED_LED_PIN 10
#define GREEN_LED_PIN 11
#define BLUE_LED_PIN 12
#define BUZZER_PIN 9
// Thresholds for gas levels
#define SAFE_THRESHOLD 400
#define CAUTION_THRESHOLD 800
void setup() {
    // Initialize serial communication
    Serial.begin(9600);
    // Initialize GPIO pins
    pinMode(RED_LED_PIN, OUTPUT);
    pinMode(GREEN_LED_PIN, OUTPUT);
    pinMode(BLUE_LED_PIN, OUTPUT);
    pinMode(BUZZER_PIN, OUTPUT);
}
void loop() {
    // Read analog value from gas sensor
    int gasValue = analogRead(GAS_SENSOR_PIN);
    // Print gas value to serial monitor
    Serial.print("Gas Value: ");
    Serial.println(gasValue);
    // Check gas concentration level and trigger LED and buzzer accordingly
    if (gasValue < SAFE_THRESHOLD) {
        // Safe level, turn on green LED
        digitalWrite(RED_LED_PIN, LOW);
        digitalWrite(GREEN_LED_PIN, HIGH);
        digitalWrite(BLUE_LED_PIN, LOW);
        tone(BUZZER_PIN, 0);
    }
    else if (gasValue < CAUTION_THRESHOLD) {
        // Caution level, turn on blue LED
        digitalWrite(RED_LED_PIN, LOW);
        digitalWrite(GREEN_LED_PIN, LOW);
        digitalWrite(BLUE_LED_PIN, HIGH);
        tone(BUZZER_PIN, 1000); // 1kHz tone
        delay(500);
        tone(BUZZER_PIN, 1000); // 1kHz tone
        delay(500);
        tone(BUZZER_PIN, 0);
    }
    else if (gasValue > CAUTION_THRESHOLD) {
        // Danger level, turn on red LED
        digitalWrite(RED_LED_PIN, HIGH);
        digitalWrite(GREEN_LED_PIN, LOW);
        digitalWrite(BLUE_LED_PIN, LOW);
        tone(BUZZER_PIN, 2000); // 2kHz tone
        delay(500);
        tone(BUZZER_PIN, 2000); // 2kHz tone
        delay(500);
        tone(BUZZER_PIN, 0);
    }
}
```

```
}  
// Delay before next reading  
delay(1000); // 1 second delay  
}
```

Conclusion:

In this experiment, an LPG gas detection system was successfully designed and implemented using an **MQ-6 gas sensor** and a **Raspberry Pi**. The system effectively monitored gas concentration levels in the environment and provided appropriate alerts through LEDs and a buzzer based on predefined safety thresholds.

Expected Oral Questions:

1. What is the working principle of the MQ-6 gas sensor?
2. Why is it important to detect LPG gas leaks?
3. What is the role of Raspberry Pi in this project?
4. Why do we use LEDs and a buzzer in this system?
5. What is the difference between analog and digital output of the MQ-6 sensor?

FAQs in Interview:

1. What is the function of the MQ-6 sensor in your project?
2. How does the Raspberry Pi interact with the gas sensor and other components?
3. What logic did you use to determine safe, caution, and danger gas levels?
4. How is the buzzer used to indicate different gas concentration levels?
5. What improvements would you suggest for making this system more efficient or reliable?